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Design Framework and Laboratory Experiments for Helix and Slinky Type Ground Source Heat Exchangers for Retrofitting Projects

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Abstract: The focus of the experimental work was on shallow spiral geothermal heat exchanger configurations. Real-scale experiments were carried out for vertically oriented spiral collectors (helix) in sand and soil. One objective was to develop a measurement concept in laboratory environment to create a framework for a validated database. This database serves as the basis for further and new development of engineering design tools. To achieve the highest possible data-point density in the observed environment, temperature sensors and a fiber-optic temperature measurement system (DTS) were used. Soil probes were taken in situ before and after the measurements and analyzed at a thermophysical laboratory to determine material properties. The heat flow was controlled by an electric heating cable, which was installed in the form of a spiral-shaped heat exchanger in a 1 m^3 container. To guarantee constant boundary conditions, the measurements were carried out in a climate chamber at a defined ambient temperature. The evaluation of the transient response behavior is spatially resolved. The results are coordinate-based temperature points, which describe temperature gradients in all axes of the container over time, which are combined with known soil properties. The collected data was used to develop computational fluid dynamic (CFD) models, which are used to extend the variety of geometry and soil configurations for developing new design tools.

Keywords: shallow geothermal; GHEX design; slinky-loop collectors; spiral; earth basket collector; laboratory experiments; fiber optics; DTS; CFD modeling; engineering tool; design framework; heat pumps; building retrofitting; Horizon 2020

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1. Introduction

Within the EU program "Horizon 2020", the project "GEOFIT" was started in May 2018. GEOFIT is an international project about geothermal retrofitting of buildings, co-founded by the EU Commission in the framework of LCE-17-2017—"Easier to install and more efficient geothermal systems for retrofitting buildings". Geothermal systems have been considered some of the most efficient and renewable technologies to attain sustainability goals for existing buildings. In this line, the H2020-project GEOFIT does not only consider the main geothermal concepts, such as heat exchangers and ground source heat pumps, but also their integration with heating and cooling components. GEOFIT defines heating and cooling (H/C) system retrofitting as long-term upgrading of existing building performance by increasing energy efficiency, decreasing overall energy demand, and providing comfortable indoor environments for occupants. GEOFIT will integrate these three main concepts to define novel and cost-efficient geothermal systems and will deploy an integrated constructive process to make the application of enhanced geothermal systems (EGS) in energy building retrofitting projects feasible, thus enabling the uptake of

geothermal energy as a key renewable energy source for the building retrofitting market [1]. When undertaking building retrofitting, low-intervention and noninvasive techniques are commonly required. The limited space available, and the risk of building instabilities and structural damages due to drilling or geological limitations demand specific types of HEX configurations, in general with low-depth drilling. GEOFIT will optimize a novel generation of vertical and horizontal closed-loop systems, aimed to be installed at a low depth—a maximum of five meters. These systems will be optimized by the project team and integrated with other enhanced geothermal system components like the heat pump, allowing flexibility in the installation procedure. Horizontal closed-loop heat exchangers will be used, suitable for the use of trenchless drilling approaches performed by a comprehensive drilling plan (CDP). This allows the installation of horizontal heat exchangers with minimum disturbance, smaller diameters, and the ability to make use of the limited geothermal potentials associated with existing buildings and infrastructure.

Design tools and software for GHEX, such as Earth Energy Designer EED, Ground Loop Heat Exchanger Design Software GHLEPRO, GLD, etc., exist for planning geothermal systems and design of vertical borehole heat exchanger types. Earth basket and slinky type GHEX are not common or not fully parameterizable in these tools regarding all thermodynamic parameters. To assist in develop such a design tool for slinky and earth basket HEX, experimental data describing the thermal characteristics are investigated in this paper.

The main type of ground source heat pump systems (GSHP) in Europe uses vertical borehole heat exchangers (BHE). The installation depths of these types range between 100 and 300 m, where the undisturbed ground temperature offers a robust heat source for heat pumps for up to 50 years and longer. Currently, this is a proven technology, used for more than 30 years in hundreds of thousands of systems worldwide [2].

Between 2006 and 2018, approximately 100,000 GSHP systems were installed annually throughout Europe. Although the annual number is stable, the average capacity is increasing. Applications are becoming more commercial and industrial [3].

GSHP systems use earth-coupled, ground-source, and water-source heat exchanger types. These types are either used in open- or closed-loop concepts. Closed-loop GSHP systems consist of one or more heat pumps, a ground heat exchanger, circulation pumps and a heating or cooling system to distribute the energy where it is needed. In this paper, only closed-loop systems and near-surface GHEX types up to 5 m depth are discussed. Figure 1 shows the shares of installed GHEX types from the years 2010–2018. The statistics show only borehole GHEX; therefore, horizontal flat collectors are excluded. Only 13% are the spiral GHEX type.

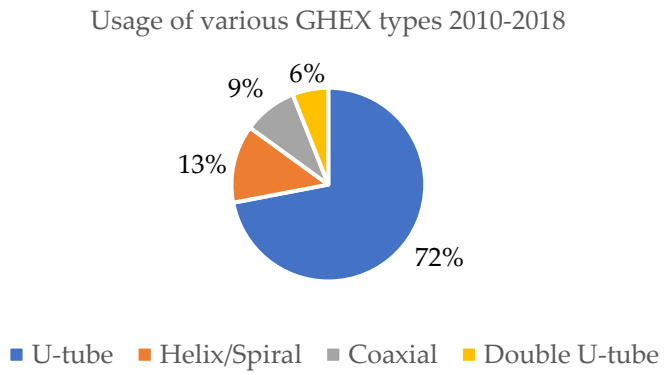


Figure 1. Usage of various GHEX types, numbers adapted from [4].

In retrofit projects, the installation of vertical borehole heat exchangers can prove difficult or impossible due to the prerequisites of the site. The typical depth of 100–200 m of vertical BHE can be a limiting factor, especially in dense urban environments, due to the limited space available for BHE fields or due to the existence of shallow groundwater bodies, which are not to be impaired by the geothermal energy use. Furthermore, existing

underground city service infrastructures such as utilities, sewage systems, and foundations must be taken into consideration and increase the complexity of logistics and the installation of vertical BHE.

The drilling depth may also be limited by law, regulations, or due to unfavorable underground conditions. Certain areas of the Netherlands, for example, are restricted. It is not permitted to drill more than 30 m below the ground surface in southern regions.

For the design of a borehole GSHP system, knowledge of the underground thermal properties, such as effective heat transfer capacity and—if applicable—the existence of groundwater bodies and the subsequent groundwater flow, is essential. In small systems for residential buildings, for example, these parameters are usually estimated. For larger systems such as commercial GSHP or borehole thermal energy storages (BTES) the thermal conductivity is measured in situ with thermal response tests. The development of the thermal response test (TRT) or geothermal response test (GRT) started in the mid-1990s. The TRT determines the effective thermal conductivity of the underground and the borehole filling. The result represents the total heat transport λ_{eff} in the underground, including conductivity and convection in permeable layers with groundwater [5] Current testing methods for borehole heat exchangers are available in ISO 17628:2015 [6].

Several design tools such as Earth Energy Designer and GLHEPRO are available on the market to calculate GSHP systems with borehole heat exchangers [7]. Therefore, one of the limiting factors is the nonexistence of standardized TRT methods specifically designed for nonstandard, highly flexible and complex heat exchanger configurations such as slinky and earth basket types. The lack of specific TRT methods makes contractors and installers choose other types of heat exchangers with standardized TRT methods available. The use of nonstandard configurations is an extra cost since each time a specific TRT method must be designed. A universal design tool for horizontal and vertical spiral GHEX has not been reported yet [8].

After own research, no relevant recent information could be found in this regard either. In recent years there have been several investigations and studies on the thermal behavior of spiral GHEX. This includes numerical, experimental, analytical, and optimizing methods. As shown in Figure 2, 50% of all studies were performed numerically with simulation software. All methods consider geometry, material, operating fluid, and installation depth in the soil. The aim of this paper is to create the basis for a method to dimension vertical spiral collectors more precisely for design purposes. The difference to the previous studies found during this research is that no hydraulic collector model with optional heat pump is used, but an electric heating cable. In addition, the material properties of the soil types are analyzed in more detail. The combination provides a more rapid and accurate simulation by means of CFD.

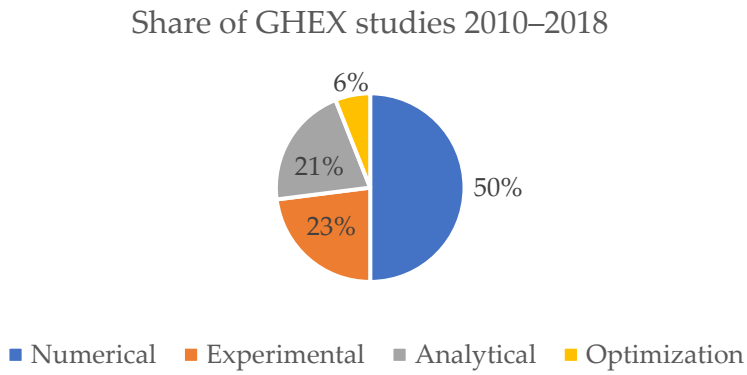


Figure 2. Share of GHEX studies by type, numbers adapted from [4].

In this paper, only vertical spiral GHEX (helix) are discussed and this type was used in the laboratory experiments.

1.1. Shallow Heat Exchanger Types

The following Table 1 shows an overview of common spiral GHEX types during installation and their typical application parameters.

Table 1. Shallow heat exchanger types.

Vertical Spiral/Helix [9]	**Horizontal Spiral [9]**	**Horizontal Slinky-Loop [10]**	**Vertical Slinky-Loop [10]**
Installation in vertical boreholes up to 5 m depth and a coil diameter of 0.2–0.5 m.	Installation in trenches up to 5 m depth and a coil diameter of 0.2–05 m.	Installation in trenches up to 2 m width and 1.5 m depth. The overlapping of the loops can range between no overlap and half of the diameter.	Installation in trenches about 0.5 m width
			

1.2. Tools Available for GHEX Design

An overview of design tools currently available in which spiral GHEX-type can be modeled was produced. As can be seen from Table 2, the functions, thermodynamic processes, and features integrated differ.

List of compared software:

EED: Earth Energy Designer, BLOCON AB, https://buildingphysics.com (accessed on 10 April 2022)
GHLEPRO: Ground Loop Heat Exchanger Design Software, https://hvac.okstate.edu (accessed on 10 April 2022)
EWS: ErdWärmeSonden, Huber Energietechnik AG, EWS | Huber Energietechnik AG, Zurich (hetag.ch) accessed on 10 April 2022)
GLD: Ground Loop Design, Thermal Dynamics, www.groundloopdesign.com (accessed on 10 April 2022)
FEFLOW: Finite Element subsurface FLOW system, DHI, www.feflow.info (accessed on 10 April 2022)
SBM: Superposition Borehole Model for TRNSYS
DST: Duct Storage Model for TRNSYS
COMSOL: Multiphysics Simulation Software, COMSOL, www.comsol.de (accessed on 10 April 2022)

Table 2. Comparison of GHEX design tools (AIT, 2019).

Design Tool	EED	GHLEPRO	EWS	GLD	FEFLOW	SBM	DST	COMSOL
Analytical g-function	X	X	-	X	-	-	-	-
Numerical g-function	-	-	X	-	X	X	X	X
Arbitrary time step	-	-	-	?	X	X	X	X
Conductive heat flow	X	X	X	X	X	X	X	X
Advective heat flow	-	-	-	-	X	-	-	X
Layered geology	-	-	-	-	X	-	partly	X
Arbitrary topology	-	-	-	X	X	X	-	X

G-functions:

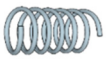
This function represents the thermal response factors and is the most important information of a GHEX. Depending on the model, it describes all heat flows in the existing media such as fluid, pipe wall, and soil, as far as possible. These g-functions can either be modelled analytically, semi-analytically, or numerically. Analytical methods are the infinite line-source model (ILS), cylindrical heat-source model (CHS), and finite line-source model (FLS), which are well-described in literature. Based on these models, ground-heat transfer can be predicted [11].

In recent years, several studies have been investigated experimentally as well as numerically. An overview and some key findings of the thermal performance of spiral GHEX are given in the paper "Numerical Simulation of a Novel Spiral-Type Ground Heat Exchanger for Enhancing Heat Transfer Performance of a Geothermal Heat Pump" [12]. In contrast to this paper, the listed investigations include hydraulic systems in combination with heat pumps and simulations with COMSOL. Furthermore, the influence of fluid behavior within the pipes, changes in pitch, and the number and spacing of several spiral collectors are dealt with.

2. Materials and Methods

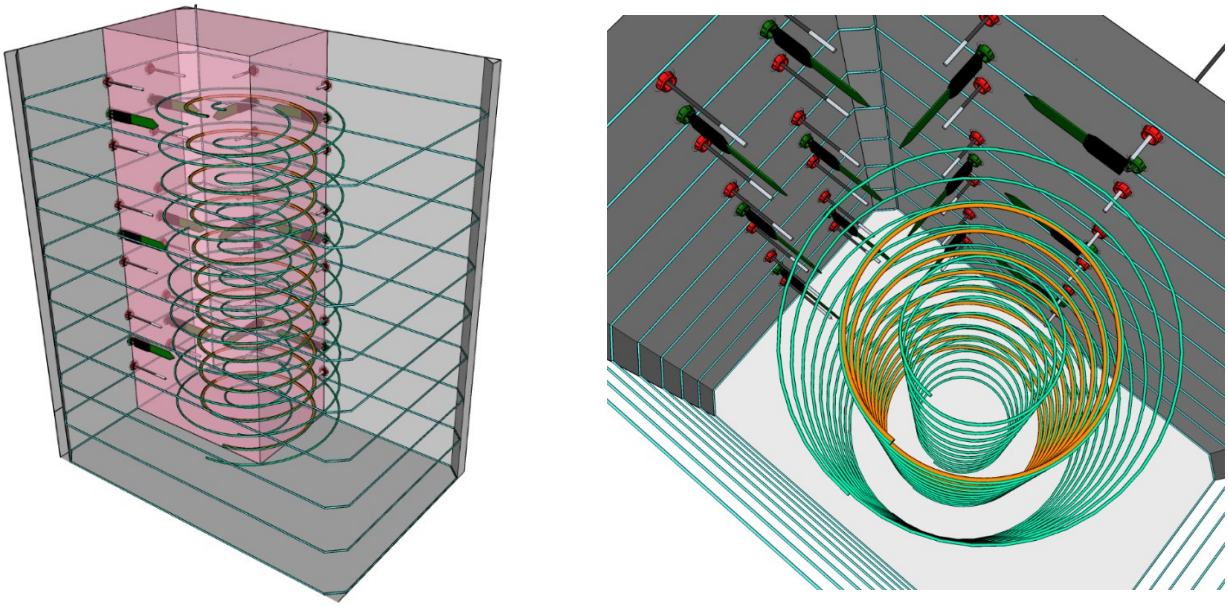
For the creation of an experimental database, the following combinations of GHEX systems were defined (Table 3) within the research project, whereas this paper comprises the work on one of these variants, namely vertical helix, sand, soil, and heat injection mode. A total of four different geometries in three soil types were supposed to be measured. Within these basic variants, the geometry like loop pitch, diameter, as well as different soil temperatures were supposed to be tested. The use of an electric heating cable facilitates the control of energy input. Electrical energy can be controlled more precisely and faster than hydraulic piping system. Based on the data acquired with these measurements, CFD simulations are to be developed. Since experimental tests are time-consuming and cost-intensive, validated CFD simulations offer a wide range of variations in terms of material parameters and geometries.

Table 3. Experimental overview.

	Vertical Helix	**Horizontal Helix**	**Horizontal Flat-Slinky**	**Vertical Flat Helix**
Geometry Types				
Soil types	Gravel/soil/mixture Sand/Soil/50/50			
Heat flow rates	Heat extraction	**Heat injection**		

2.1. Materials

The experiment was designed to be set up in a 1.1 m^3 HD-PE container (L × W × H 1130 mm × 725 mm × 1350 mm), which was situated in a climatic chamber. Constant soil parameters can be realised as boundary conditions via the climate chamber controlling. A HD-PE plastic grid net with a mesh size of 30 mm × 25 mm and a total height of 1200 mm acts as a support structure for the geometry of the heating cable (not depicted in Figure 3a,b). To measure temperature gradients in all axes (XYZ) and to determine the moisture content of the soil, a measuring setup based on PT1000 sensors, FDR-based moisture sensors, and a DTS system was designed. Due to symmetry, only one quarter, shown in pink in Figure 3a, was monitored. To reduce the influence of the sensors on the soil concerning thermal conductivity, a DTS system (Sensortran 5100A) was used. Such systems are often used for vertical boreholes and in the oil industry for monitoring piping networks. DTS systems provide a temperature profile along a fiber-optic cable. This has the advantage of a higher density of data points and not having to install individual sensors, including supply lines.



(**a**) (**b**)

Figure 3. Rendering of the container experiment setup. Isometric side view (**a**) and top view (**b**).

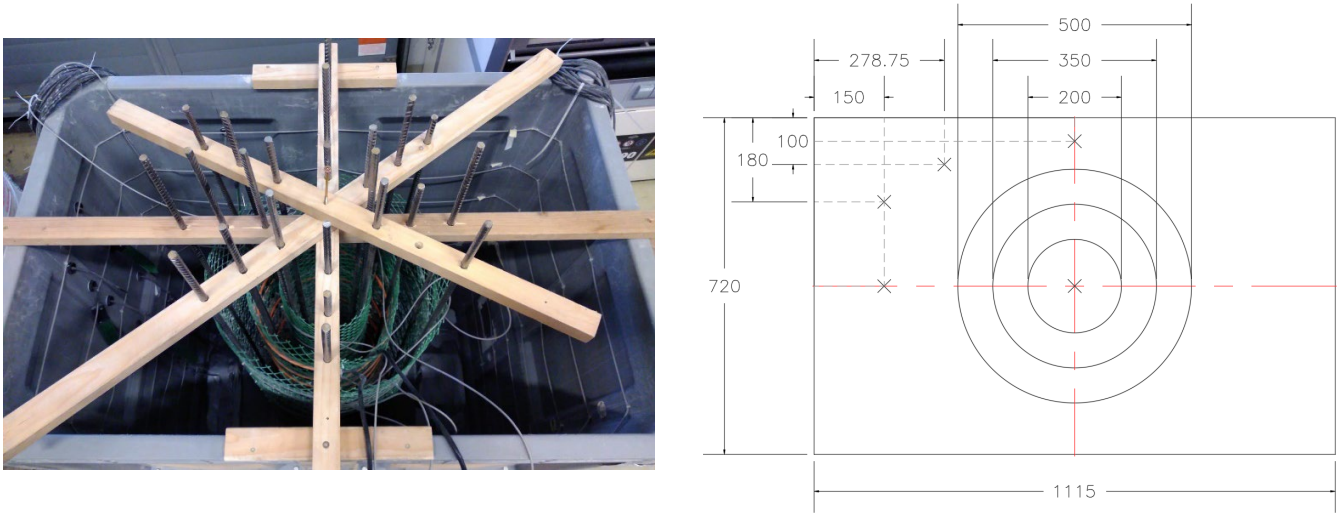
Figure 4 shows how the sensors were aligned during the filling of the container. On the left side (Figure 4a) two PT1000 sensors and on the right side (Figure 4b) moisture sensors SMT100 are visible.



(**a**) (**b**)

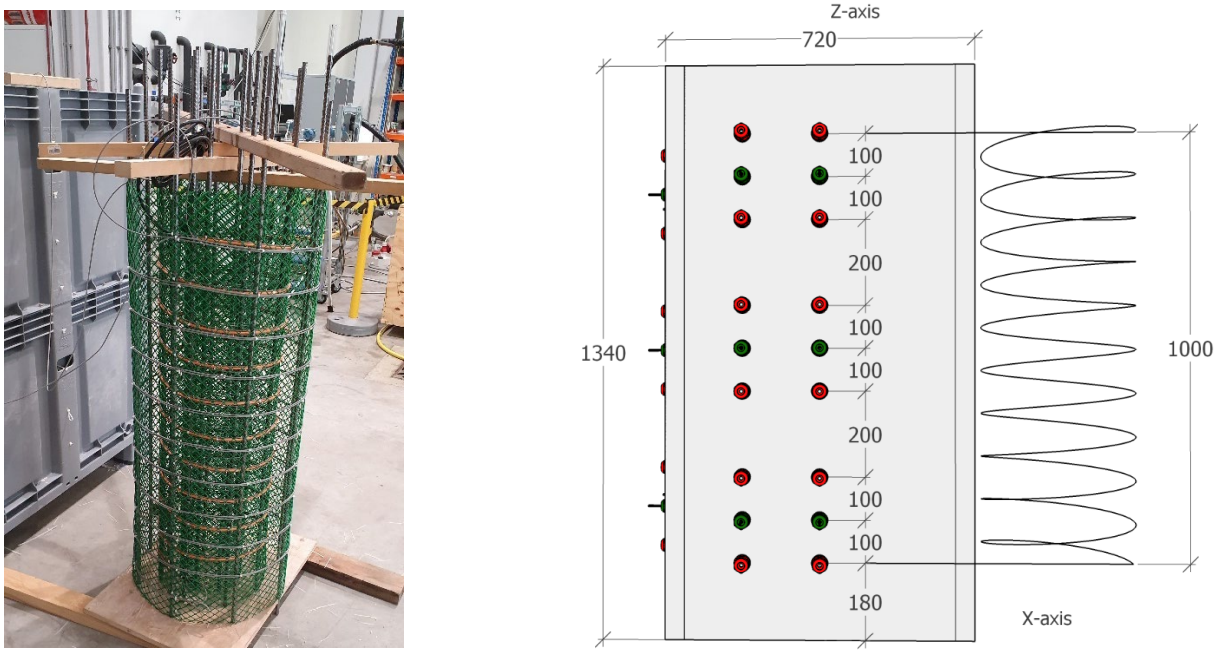
Figure 4. Figure **4.** Laboratory experimental setup during the buildup process. PT1000 (**a**) and SMT100 (**b**).

Figure 5 shows all dimensions in the container top view with the diameters of the baskets as well as the sensor installation depths in X and Y directions. The positions marked with "X" (b) represent the sensor measuring points. Figure 6b shows the side view to depict the sensor positions in the Z-direction. In addition, the location of the helix is shown. Red markings show PT1000 and green markings show SMT100 sensors. On the left side (a) the physical experiment model is shown with its supporting structure. After the filling of the container, the iron rods and the wooden centric bars were removed.



(**a**) (**b**)

Figure 5. Experimental setup top view (**a**) and all dimensions and positions of sensors (**b**) in mm.



(**a**) (**b**)

Figure 6. (**a**) physical model and (**b**) 2D dimensions of the container and sensor positions side view (dimensions in mm).

2.2. Methods

2.2.1. Data Acquisition

A central system based on LabView was developed for control and data acquisition (Figure 7). There, all signals of the sensors except the DTS can be processed and analog in- and outputs can be controlled. PT1000 and SMT100 moisture sensors were used to monitor the data inside of a quarter of the container. A DTS fiber-optic cable was mounted next to the heating cable, in two different soil layers parallel in- and outside to the helix and next to the container inner surface to define a boundary condition. Raw data of all sensors and temperature and moisture values were logged. The heating cable was regulated by a digital SCR voltage controller. For monitoring and measuring the electric parameters as voltage, current, and power an Agilent 34970A connected to LabView was used.

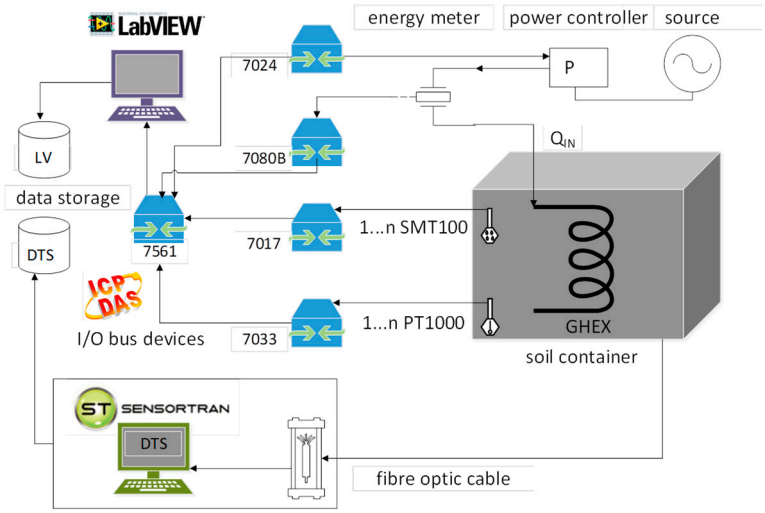


Figure 7. Data acquisition concept.

Figure 8 depicts the defined coordinate system inside the container related to the DTS fiber-optic cable. The datapoint resolution per meter of fiber-optic cable is 2 (2/m). To double the amount of data points to 4/m, the fiber-optic cable was installed bi-directionally. This results on a data point every 0.25 m.

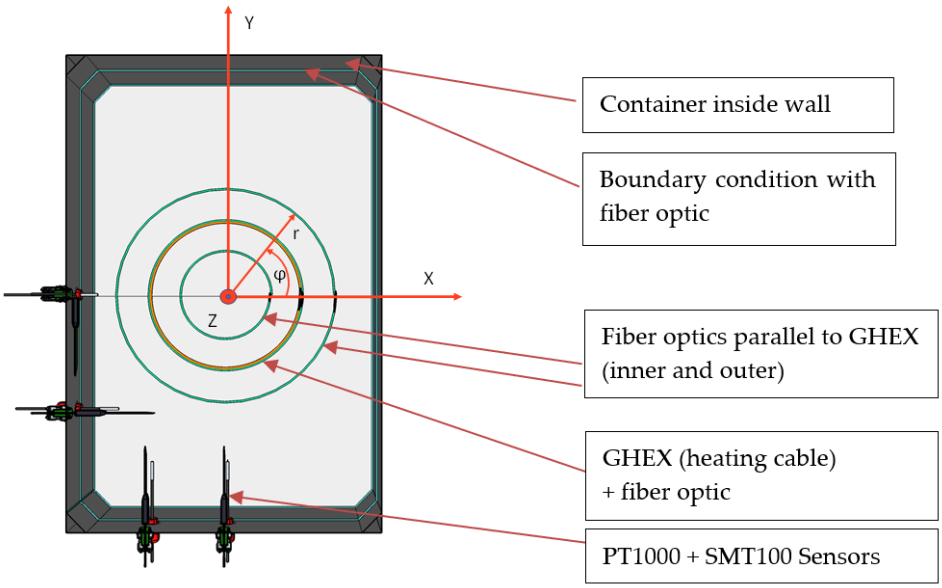


Figure 8. Coordinate system in container—isometric top view.

Figure 9 shows the data points of the DTS system based on the three different diameters. "Basket" refers to the respective support grid structures where the DTS cable is installed in the same pitch level as the heating cable. The innermost basket (blue) has the smallest number of data points. The basket with the heating cable (middle basket) is shown in orange, and the outer basket, shown in green, has the largest diameter.

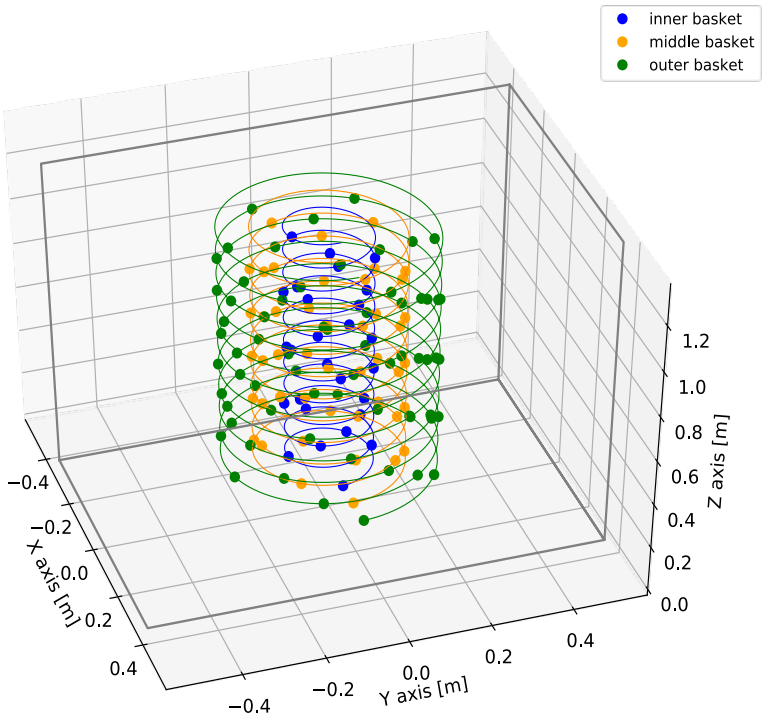


Figure 9. DTS data points.

2.2.2. Material Properties

The laboratory for thermophysics and thermal analysis at AIT is capable of the measurement of thermophysical properties such as specific heat capacity, as well as effective thermal conductivity of the used materials—sand, soil, and mixtures at different moisture levels. Additionally, the actual moisture content of material samples was evaluated by drying experiments. The specific heat capacity and effective thermal conductivity of the different materials were determined by a heat flow meter (HFM)—NETZSCH 446 Figure 10b). The HFM technique is a steady-state method of measuring the effective thermal conductivity of solid materials by applying a defined temperature difference across the sample (Figure 10c) and measurement of the heat flow over a defined area resulting in the heat flux density. The specific capacity of low-conductive materials can be measured with the step-heat method. This involves increasing the temperatures of both top and bottom plates (Figure 10a) simultaneously, while keeping the temperature difference between the plates at a minimum. The obtained "zero delta T" control is crucial for accurate measurements. As the temperature of both plates increases from a stable value to the desired upper set-point, the heat-flux signals are collected and integrated over time, thereby providing the total heat entering the specimen during each step. The moisture content and actual mass loss due to the drying of the materials was measured with a thermogravimetric analyzer (TGA)—NETZSCH STA 449 F1. Samples were heated up to 105 °C until no further mass loss was detected. The determined mass loss is equated with the water content of the sample.

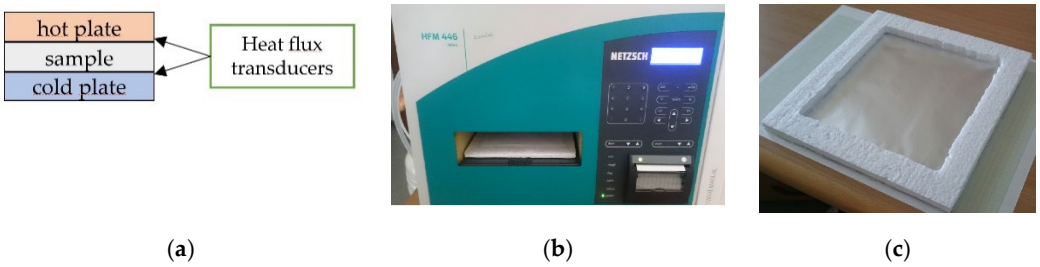


Figure 10. (**a**) HFM measuring principle; (**b**) measurement device; (**c**) sample holder frame.

In this report period, the effective thermal conductivity of the sand and the soil, as well as a mixture of both in dry and defined moist state was determined. Figure 11 depicts an example of moist (a) and dry (b) sand during three measurement cycles to determine the mean value of effective thermal conductivity (λ).

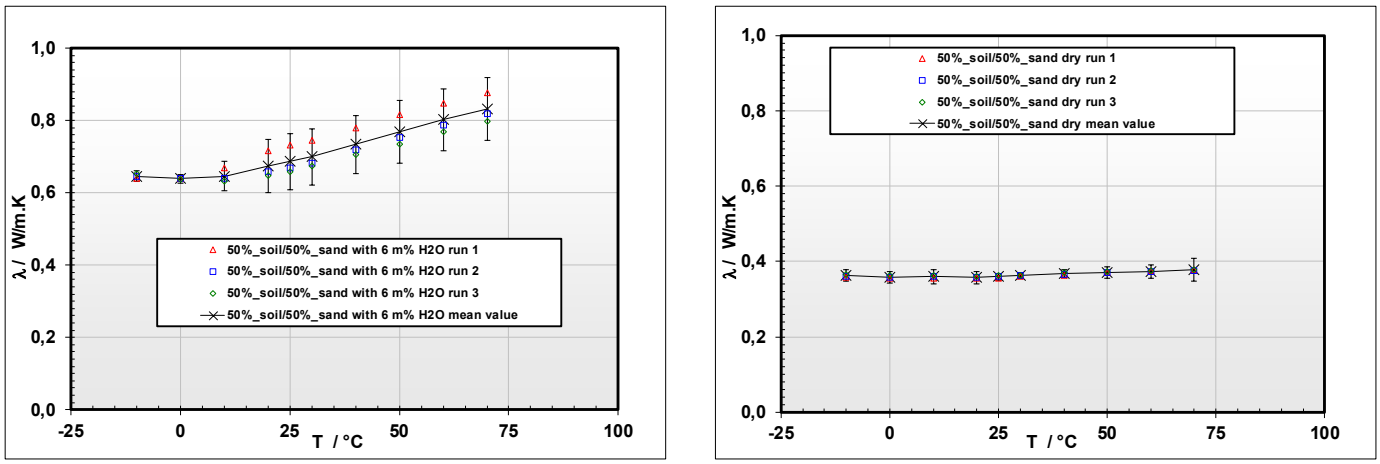


Figure 11. Effective thermal conductivity of a 50/50 soil/sand mixture with 6% water and as a dry mixture.

3. Results

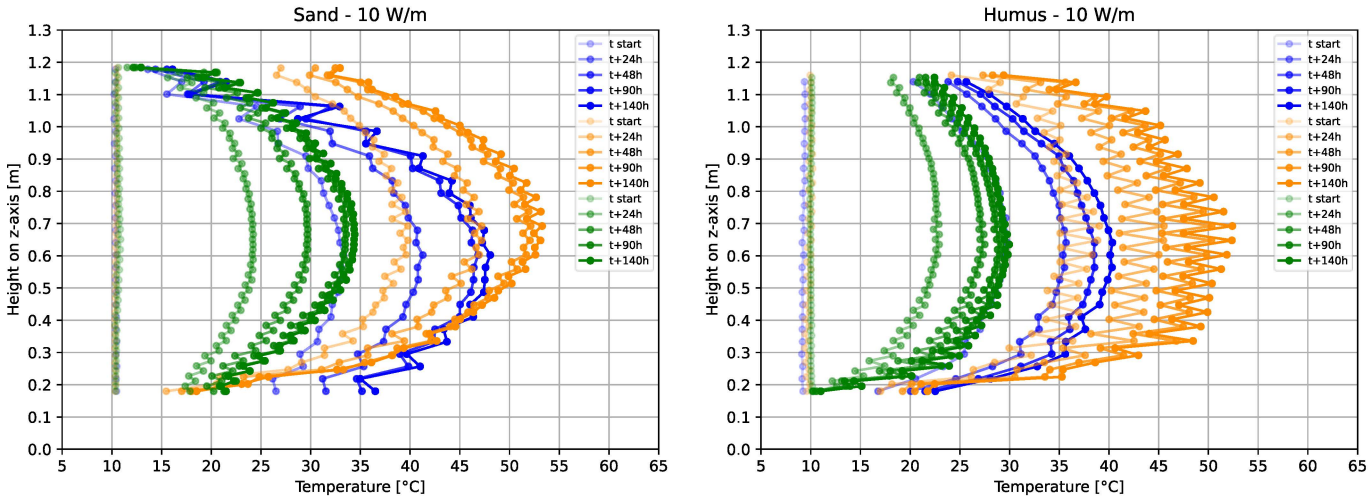
In the following paragraphs, selected experimental results are presented and compared. For comparison, a measurement with sand and soil with a specific heating power of the spiral collector (helix) of 10 W/m are used in each case. In combination with determined material data, these provided the basis for the development of CFD simulations that are not discussed in this paper.

3.1. Experimental

Figure 11 shows results based on the following experiment settings:

- Geometry of helix: h = 1 m, d = 0.35 m, p = 0.1 m
- Specific heating rate: 10 W/m–113.1 W total
- Soil type: Sand (a) and soil (b); untreated H_2O content
- Ambient temperature: 10 °C as starting condition

Figure 12 shows the comparison between sand (a) and soil (b) for a measurement duration of 140 h. The aim was to achieve a quasi-stationary state in the climate chamber in order to later model the CFD calculations in a stationary state also. Due to the duration of the measurement, this was a challenge and proved to be impossible. The three colored curves represent the bidirectional DTS sensor lines per basket. In correlation to Figure 9, blue represents the inner basket (d = 0.2 m), orange the helix heating cable (d = 0.35 m), and green the outer basket (d = 0.5 m). The y-axis depicts the height in the container and the x-axis the temperature of all DTS data points. Additionally, the temperature curves from the start time in 24 h steps to the end time of 140 h are shown. The DTS points of the boundary condition (inside container wall) are not shown, as these represent the temperature of 10 °C approximately constantly over the entire measurement.



(**a**) (**b**)

Figure 12. Temporal temperature profile over height in z-axis with sand (**a**) and soil (**b**).

Figure 13 shows all temperature data points (DTS + PT100) in the form of a scatterplot at time t = 140 h. As expected, the temperature distribution reached its maximum in the center. On the left (Figure 13a) the experiment with dry sand is shown, on the right (Figure 13b) the one with untreated soil. Due to the lower heat conductivity properties of the dry sand, temperatures in the center are slightly higher than in the moist soil.

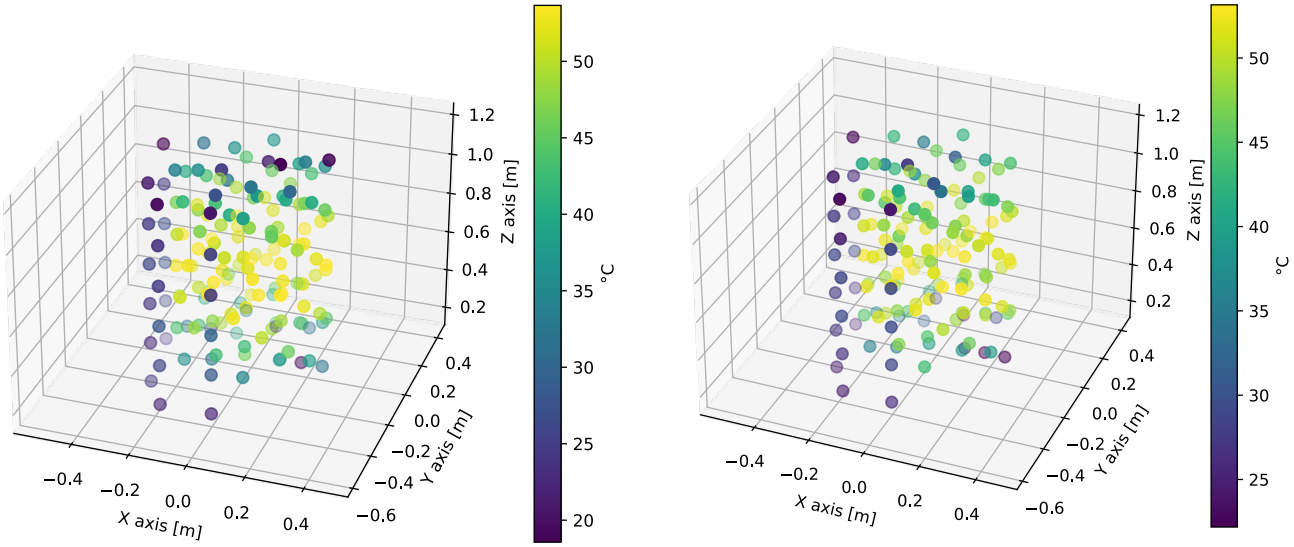


Figure 13. 3D scatter plots of all temperature data points DTS + PT1000; sand (**a**) and soil (**b**).

The humidity sensors (SMT100) did not provide any useful data during the experiments, as the values fluctuated strongly over the entire series of measurements, most likely due to a calibration error. To determine the moisture content of the soil, which has a significant influence on the material data and thus the results, material samples were taken in situ at various positions before and after the experiment. These were then measured using the mass thermogravimetry method.

3.2. Material Data

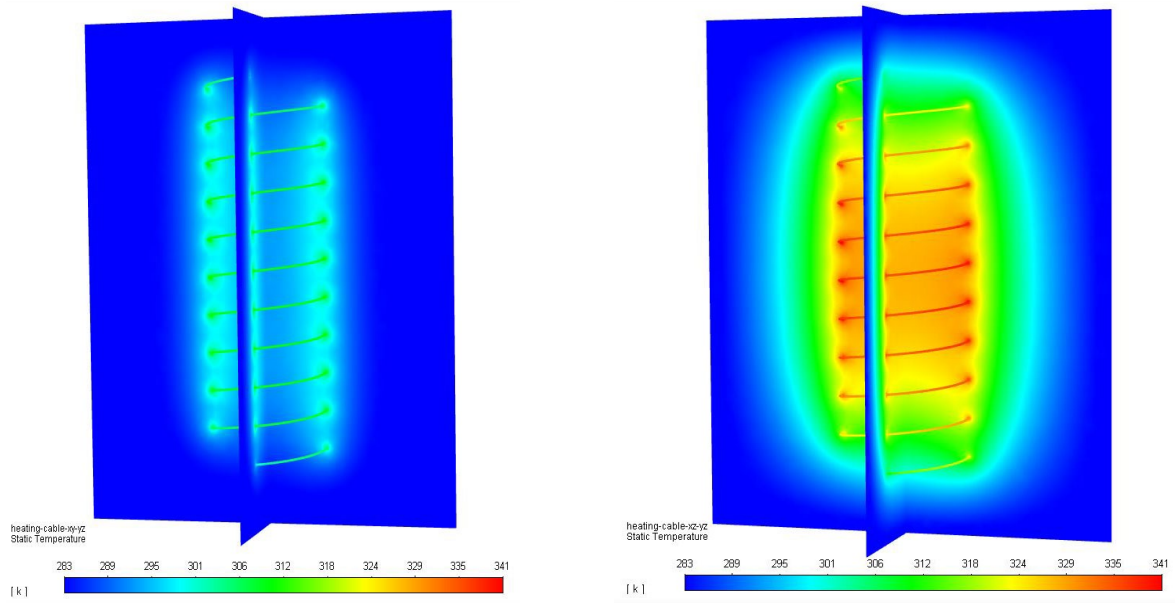
The material data of the materials used were determined using thermophysical methods, as described in 2.2.2. The following Table 4 shows the material parameters for a temperature range from −10 to +70 °C. The range was adjusted to the starting temperature of 10 °C and the final temperature in the warmest zone was no more than 55 °C. Although the sand was used untreated, the moisture content was between 0.1 and 0.5%, which can be described as quasi-dry. The materials were used untreated. Homogeneous humidification of 1 m^3 of soil would not have been feasible in this form.

Table 4. Material properties.

	Sand						**Soil**					
	Dry < 0.1%			**Moisture 6.14%**			**Dry < 0.1%**			**Moisture 6.24%**		
T	**cp**	**λ**	**ρ**	**cp**	**λ**	**ρ**	**cp**	**λ**	**ρ**	**cp**	**λ**	**ρ**
°C	J/kgK	W/mK	kg/m^3	J/kgK	W/mK	kg/m^3	J/kgK	W/mK	kg/m^3	J/kgK	W/mK	kg/m^3
−10	0.889	0.3809	1822	1.070	1.2435	1928	0.913	0.3401	1942	n.a.	0.504	2004
0	n.a.	0.3784	1822	n.a.	0.9788	1928	0.966	0.3370	1942	n.a.	0.490	2004
10	0.947	0.3796	1822	1.095	0.9514	1928	0.974	0.3251	1942	n.a.	0.483	2004
20	0.941	0.3840	1822	1.152	0.9339	1928	1.006	0.3162	1942	n.a.	0.502	2004
25	0.962	0.3884	1822	1.100	0.9619	1928	n.a.	0.3121	1942	n.a.	0.509	2004
30	0.985	0.3874	1822	1.116	0.9728	1928	1.056	0.3068	1942	n.a.	0.515	2004
40	1.027	0.3896	1822	1.134	1.09	1928	1.086	0.2991	1942	n.a.	0.536	2004
50	1.065	0.3915	1822	1.139	1.19	1928	1.085	0.3058	1942	n.a.	0.552	2004
60	1.038	0.3929	1822	1.184	1.31	1928	1.072	0.3024	1942	n.a.	0.562	2004
70	1.061	0.3873	1822	1.295	1.46	1928	1.106	0.2972	1942	n.a.	0.566	2004

3.3. CFD

Parallel to the experimental measurements, a CFD model was developed. The geometry and properties of the heating cable were defined exactly according to the physical test setup. The geometry model, from which the simulation mesh is created, was developed in form of a parameter construction in ANSYS Workbench ©. Thus, changes in the geometry can be made efficiently later to study the behavior of different helix pitches, helix diameters, and box geometries in detail. Figure 14 shows the result of a transient simulation at start and end time during the injection of energy into the system.



(**a**) (**b**)

Figure 14. CFD simulation of the helix at t_start (**a**) and t_140h (**b**).

4. Discussion

During the first measurements with the DTS, unexpected problems occurred, which were explained by the length of the fiber-optic cables being too short, which caused unstable measurement conditions. The 30-m cable used caused noise signals during operation, therefore, the setup had to be modified. Instead of multiple short lines, a single 200-m fiber-optic cable was used. This was an important finding during this experiment.

Furthermore, it was an important task to assign the DTS measuring data points on the real physical cable. Especially in this application, it was of high relevance to know the exact position of the measuring point in the XYZ direction. It was found that the number of measuring points per meter (2/m) is not suitable, as relatively few data points were available in the inner basket due to the diameter. To improve the datapoint density, the cable was installed bidirectional to increase the datapoints per meter to 4/m.

Thermal properties such as thermal conductivity and specific heat capacity increase by the absorption or mixing of the soil with water. Therefore, the exact determination of the material properties is of high relevance.

The main objective of developing and establishing a laboratory measurement and analysis method for such a GHEX model was successfully implemented. The setup is modular and can be easily extended and modified for further experiments and measurements. The data sets can be integrated into simulation software.

CFD simulations are challenging due to the complex geometry of the spiral-shaped ground collectors and the numerous influencing parameters such as heat and phase transitions of the ground due to freezing. The approach adopted within the GEOFIT project is to develop models with a high level of spatial and temporal detail and a high order of

complexity to study different key aspects of the heat-transfer problem. These models will consider the local and global problem. The local problem is described as the energy flow between the GHEX and its surrounding ground and the global problem is the heat flow between adjacent GHEX and or groundwater flow effects.

5. Summary and Conclusions

In order to characterize shallow GHEX, experimental tests were carried out in a laboratory environment. The aim was the measurement of a vertical helix (spiral) GHEX in a container of 1 m^3 filled with sand and soil. The implementation was realized by developing a measurement concept. This concept includes the measurement of temperature and moisture content of the sediment in different layers to record all temperature gradients over a certain time. In addition, a data evaluation script was programmed, which stores all data in a database and allows a detailed numerical and graphical evaluation at any time of the measurements. To simulate constant boundary conditions or undisturbed soil, the tests were carried out in a climatic chamber at constant environmental parameters. A total of seven measurements were carried out. Only three of them resulted in usable data sets due to technical problems on the power controller and the DTS system. Two of them are shown in this paper.

A combination of experimental results and CFD simulations will be used to characterize various complex geometries of GHEX [13]. The results can be used for the improvement or new development of an engineering tool for the design and dimensioning of spiral GHEX in combination with the heat pump system, since only a few practical and reliable calculation exist to date. The design of a new engineering tool for spiral GHEX will help to increase the share of these collector types, which are especially suitable for geothermal retrofitting.

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